GOALOS MISSION OS

GoalOS Mission OS

The Proof OS product package for autonomous AI work that runs until proof is done.

What this pack gives you

This pack turns the GoalOS Mission OS thesis into commercial, technical, and launch-ready assets. It is designed for a nontechnical operator to open one folder, understand the product, launch the first public-facing page, give Codex a precise implementation task, and run an until-done Evidence Docket workflow safely.

Set the objective. GoalOS runs until proof is done.

File	Use first when you want...
01 Whitepaper	the formal theory, equations, and category-defining language
02 Product Spec	the commercial product architecture and deliverables
03 Codex Prompt	an implementation-ready prompt for the GitHub repository
04 Runbook	the GitHub Actions / operator instructions
05 Launch Copy	homepage, posts, email, and pricing language
06 Deck	investor, partner, and executive presentation
Technical Code Kit	schemas, examples, scripts, and workflows to copy into the repo

The immediate product

GoalOS Mission OS: the Proof OS for autonomous AI work.

GoalOS Mission OS should not be positioned as another autonomous research assistant. Its commercial primitive is the Governed Decision State: an inspectable package of evidence, verification, risks, action, rollback conditions, Chronicle memory, and reusable capability.

The product ladder is Mission Room, Mission OS, Evidence Vault, Verifier Mesh, Chronicle, and AGIALPHA Settlement Readiness.

The fastest safe launch path

- Publish /mission-os with the Mission OS hero and mission intake form.
- Run three public-safe example missions.
- Generate Evidence Dockets, decision states, risk ledgers, action graphs, and Chronicle entries.
- Open five founding proof-mission pilot slots.
- Automate the until-done loop through GitHub Actions, but keep publication human-gated.

GoalOS Mission OS

The Proof OS for Autonomous AI Work

Vincent Boucher
MONTREAL.AI / QUEBEC.AI

v12 – Commercial Proof-to-Action Edition

Canonical line

Set the objective. GoalOS runs until proof is done.
AI creates output. GoalOS creates proof. No proof, no settlement. No eval, no propagation. No rollback, no release.

1 Executive thesis

GoalOS Mission OS is the commercial front end of the Proof OS for AI Work. It turns a high-stakes objective into a governed decision state: evidence, verification, risk, action, rollback conditions, Chronicle memory, and reusable capability. The product is not another report generator. The product is a mission-native operating system that runs until proof is done.

2 The primitive: Governed Decision State

A report is static. A Governed Decision State is live, inspectable, and operational. It contains the mission objective, decision to support, claims matrix, source provenance, contradiction register, Evidence Docket, verifier report, risk ledger, action graph, rollback conditions, Chronicle entry, capability package, and next mission recommendation.

3 Mission value

GoalOS optimizes mission value rather than report length:

$$V_{mission} = \frac{I_{decision}\, P_{integrity}\, A_{actionability}\, R_{reuse}}{C_{cost} + R_{risk} + L_{latency} + D_{proof\ debt}}. \tag{1}$$

The highest-value output is not the longest report. It is the shortest path from uncertainty to justified action.

4 AI work equation

$$W_{AI} = O \times P \times V \times S \times R, \tag{2}$$
$$\text{where } O = \text{output}, \quad P = \text{proof}, \quad V = \text{validation}, \tag{3}$$
$$S = \text{settlement or state transition}, \quad R = \text{reuse}. \tag{4}$$

If proof is zero, institutional work is zero:

$$P = 0 \quad \Rightarrow \quad W_{trusted} = 0. \tag{5}$$

5 Mission lifecycle

$$\text{Objective} \rightarrow \text{Mission Contract} \rightarrow \text{Planning} \tag{6}$$
$$\rightarrow \text{Autonomous Work} \rightarrow \text{Verification} \tag{7}$$
$$\rightarrow \text{Evidence Docket} \rightarrow \text{Decision State} \tag{8}$$
$$\rightarrow \text{Action Graph} \rightarrow \text{Chronicle}. \tag{9}$$

6 Acceptance law

A mission artifact is accepted if and only if all mandatory gates clear:

$$Accept(m) = 1 \iff ProofBundle(m) \wedge ValidatorClear(m) \tag{10}$$
$$\wedge\, SourceProvenance(m) \wedge ContradictionsLogged(m) \tag{11}$$
$$\wedge\, RiskLedger(m) \wedge ActionGraph(m) \tag{12}$$
$$\wedge\, ClaimBoundaryPass(m) \wedge ReviewReady(m). \tag{13}$$

Otherwise, $Accept(m) = 0$.

7 Until-done autonomy

Autonomy is an artifact-generation loop, not a governance bypass. The run state is:

$$\Sigma_t = (M_t, A_t, Q_t, B_t, D_t), \tag{14}$$

where M_t is mission state, A_t generated artifacts, Q_t quality gates, B_t boundary checks, and D_t done status. The system advances by repair:

$$\Sigma_{t+1} = Repair(Generate(\Sigma_t), Q_t, B_t). \tag{15}$$

The loop halts only when:

$$DONE = 1 \iff RequiredArtifactsComplete \tag{16}$$
$$\wedge\, EvidenceComplete \wedge BoundaryPass \tag{17}$$
$$\wedge\, QAReportPass \wedge HumanReviewReady. \tag{18}$$

8 Verifier Mesh

The Verifier Mesh scores the integrity of the mission package:

$$\mathcal{V}(m) = w_s S_{sources} + w_c C_{claims} + w_x X_{contradictions} \tag{19}$$

$$+ w_r R_{risk} + w_a A_{action} + w_b B_{boundary}. \tag{20}$$

A verifier is useful only when it blocks unsupported propagation. Scores are advisory; gates are mandatory.

9 Evidence Graph moat

Let the Evidence Graph be:

$$EG_t = \{Missions, Claims, Sources, Proofs, Reviews, Risks, Actions, Outcomes, Capabilities\}_{0:t}. \tag{21}$$

The moat is the time-integrated quality of the graph:

$$Moat(t) = \int_0^t \Big(Q_{evidence}\, R_{reuse}\, X_{review}\, S_{settlement}\Big) d\tau \tag{22}$$

$$- D_{proof\ debt} - K_{copyability}. \tag{23}$$

A demo can be copied. A living evidence graph cannot be retroactively fabricated.

10 Product ladder

Layer	Commercial meaning
Mission Room	One objective becomes one proof-backed decision state.
Mission OS	Recurring mission operations across a team or organization.
Evidence Vault	System of record for missions, proofs, decisions, risks, and capability packages.
Verifier Mesh	Review, challenge, and validate claims before action.
Chronicle	Institutional memory of what proved useful.
AGIALPHA Settlement Readiness	Optional proof-settlement layer where value movement remains human-gated.

11 Claim boundary

GoalOS Mission OS does not claim achieved AGI, ASI, superintelligence, guaranteed ROI, legal advice, financial advice, medical advice, production readiness, external audit completion, cybersecurity certification, mainnet deployment, or guaranteed market outcomes. The claim is product-operational: a mission-native system can generate proof-backed decision states and package evidence for human-governed review.

12 Canonical conclusion

Research ends with a report. GoalOS ends with a governed decision state. Insight is not finished until it becomes governed action. Set the objective. GoalOS runs until proof is done.

GoalOS Mission OS: The Equations of Proof-to-Action Work

Set the objective. GoalOS runs until proof is done.

AI creates output. GoalOS creates proof. A report informs; a governed decision state can be inspected, defended, acted on, and reused.

Mission value

$$V_{mission} = \frac{I_{decision} P_{integrity} A_{actionability} R_{reuse}}{C_{cost} + R_{risk} + L_{latency} + D_{proof\ debt}}$$

AI work

$$W_{AI} = O \times P \times V \times S \times R,$$
$$P = 0 \Rightarrow W_{trusted} = 0.$$

Acceptance

$$\begin{aligned} Accept(m) = 1 \iff & ProofBundle(m) \wedge ValidatorClear(m) \\ & \wedge SourceProvenance(m) \wedge RiskLedger(m) \\ & \wedge ActionGraph(m) \wedge ClaimBoundaryPass(m). \end{aligned}$$

Until-done loop

$$\Sigma_{t+1} = Repair(Generate(\Sigma_t), Q_t, B_t),$$
$$\begin{aligned} DONE = 1 \iff & RequiredArtifactsComplete \\ & \wedge EvidenceComplete \wedge BoundaryPass \\ & \wedge QAReportPass \wedge HumanReviewReady. \end{aligned}$$

Verifier Mesh

$$\begin{aligned} \mathcal{V}(m) = & w_s S_{sources} + w_c C_{claims} + w_x X_{contradictions} \\ & + w_r R_{risk} + w_a A_{action} + w_b B_{boundary}. \end{aligned}$$

Evidence Graph

$$\begin{aligned} EG_t = \{ & Missions, Claims, Sources, Proofs, Reviews, \\ & Risks, Actions, Outcomes, Capabilities\}_{0:t}. \end{aligned}$$

Moat

$$\begin{aligned} Moat(t) = & \int_0^t (Q_{evidence} R_{reuse} X_{review} S_{settlement}) d\tau \\ & - D_{proof\ debt} - K_{copyability}. \end{aligned}$$

Operating doctrine

No proof, no settlement. No eval, no propagation. No rollback, no release.

The deliverable is not a document. The deliverable is a governed decision state.

GOALOS MISSION OS

GoalOS Mission OS Product Spec

The commercial architecture, offer ladder, and first-pilot product plan.

Category and product thesis

GoalOS Mission OS turns high-stakes objectives into proof-backed governed decision states.

The market has validated long-horizon autonomous research. GoalOS should not answer with a research assistant. It should own the layer above: autonomous Proof-to-Action work. The output is not merely a report; it is a decision state that can be inspected, defended, acted on, and reused.

The category language is Proof OS for AI Work. The product language is GoalOS Mission OS. The primitive is the Governed Decision State.

Governed Decision State

A Governed Decision State is a mission artifact that records what was asked, what was found, what supports it, what contradicts it, what risks remain, what action follows, what proof is attached, what can be reused, and what must happen next.

Component	Meaning
Mission Contract	Objective, decision to support, success criteria, constraints, data boundary, risk class, deadline, done conditio
Evidence Docket	Claims matrix, source provenance, proof packets, contradiction register, verifier report, cost/risk ledgers, and
Decision State	Options, assumptions, confidence, risks, recommendation, unresolved unknowns, falsification path.
Action Graph	Tasks, owners, dependencies, deadlines, proof required, approval gates, rollback conditions.
Chronicle Entry	Durable memory of the mission, evidence, decisions, reusable patterns, and next missions.

Mission Value equation

```
Mission Value = (Decision Impact x Proof Integrity x Actionability x Reuse) / (Cost + Risk + Latency + Proof Debt)
```

GoalOS optimizes mission value, not report length. The highest-value output is not the longest report. It is the shortest path from uncertainty to justified action.

Product tiers

Tier	Offer	Suggested packaging
Mission Room	One high-stakes mission	48-hour proof-backed decision package: Evidence Docket, Decision State, Act
Mission OS	Recurring mission operations	Multiple missions, standard templates, Evidence Vault, and reviewer workflow

Enterprise Evidence Va	System of record	Mission archive, source provenance, risk ledger, verifier results, capability libr
Sovereign / Strategic	Private deployment	Custom verifier mesh, private data boundary, governance controls, AGIALPHA

Founding pilot offer

Five founding Mission OS pilots: one objective, 48 hours, proof-backed governed decision state.

- AI Product Intelligence mission
- Enterprise AI Adoption mission
- Policy and Regulation Options mission
- Cyber Defensive Readiness mission
- Market Entry / Partnership Diligence mission

GOALOS MISSION OS

Codex Implementation Prompt

Copy-paste prompt for implementing GoalOS Mission OS in the GitHub repository.

Prompt

```
You are Codex acting as a top-tier product engineering, AI workflow, GitHub Actions, document-generation,
evidence-systems, and SaaS launch team.

Repository:
https://github.com/MontrealAI/goalos-agialpha-ascension

Mission:
Implement GoalOS Mission OS: a commercial Proof OS product that turns a user objective into a proof-backed governed
decision state.

Do not copy any external product, branding, layout, wording, or claims. Build from GoalOS, AGI ALPHA, AEP-001, Evidence
Dockets, ProofBundles, Selection Gates, Chronicle memory, and the doctrine:

AI creates output. GoalOS creates proof.
Set the objective. GoalOS runs until proof is done.
No proof, no settlement. No eval, no propagation. No rollback, no release.

Primary product:
GoalOS Mission OS

Primary tagline:
Set the objective. GoalOS runs until proof is done.

Product primitive:
Governed Decision State

Implement:

1. /mission-os landing page
Sections: Hero, Beyond Research, GoalOS Flow, Governed Decision State, Board-Ready Outputs, Why GoalOS Wins, Use Cases,
Launch a Mission.
Hero copy:
GoalOS Mission OS
Set the objective. GoalOS runs until proof is done.
Turn high-stakes objectives into proof-backed decision states: Evidence Dockets, verifier reports, action graphs, risk
ledgers, executive briefs, and Chronicle memory.
CTAs: Launch a Mission; View a Proof Bundle.

2. Mission intake schema
Create schemas/goalos-mission-os-intake.schema.json with fields: mission_title, decision_to_support, success_criteria,
failure_criteria, constraints, allowed_sources, private_data_boundary, risk_class, deadline, output_package,
reviewer_required, claim_boundary, done_condition.
```

3. Example missions
Create examples/mission-os/ai-product-intelligence.json, enterprise-ai-adoption.json, policy-regulation-options.json, cyber-defensive-readiness.json.

4. Until-done mission generator
Create scripts/mission-os/mission_os_until_done.py. Given a mission JSON, generate and repair until all required outputs pass checks:
GoalOSCommit.md, RunCommitment.md, MissionPlan.md, WorkGraph.json, ClaimsMatrix.csv, SourceProvenance.csv, ContradictionRegister.md, EvidenceDocket.md, VerifierReport.md, RiskLedger.csv, DecisionState.json, ExecutiveBrief.md, DecisionDeck.md, ActionGraph.md, ChronicleEntry.md, CapabilityPackage.md, ClaimBoundaryReport.md, QAReport.md, artifact-manifest.json, index.html.

5. Done checker
Create scripts/mission-os/done_check.py. DONE=true only if all required files exist and claim-boundary and QA reports pass.

6. Claim-boundary checker
Create scripts/mission-os/claim_boundary_check.py. Block unsupported claims including: achieved AGI, achieved ASI, superintelligence achieved, guaranteed ROI, legal advice, financial advice, medical advice, production-ready, externally audited, cybersecurity certified, mainnet deployed, guaranteed market outcome, guaranteed investment outcome.

7. GitHub Actions
Create .github/workflows/goalos-mission-os-until-done.yml. Behavior: workflow_dispatch input mission_file, run generator, run claim-boundary check, run done check, upload artifacts, open pull request with generated mission artifacts, never auto-merge, never broadcast mainnet, never move tokens, never publish unsupported claims.

8. Docs
Create docs/GOALOS_MISSION_OS_START_HERE.md, docs/GOALOS_MISSION_OS_OPERATOR_RUNBOOK.md, docs/GOALOS_MISSION_OS_CLAIM_BOUNDARY.md, docs/GOALOS_GOVERNED_DECISION_STATE.md.

9. Templates
Create templates/mission-os-homepage-copy.md, templates/mission-os-launch-post.md, templates/mission-os-sales-email.md, templates/mission-os-pricing.md.

10. Tests
Add tests for schema validation, generator completeness, claim-boundary blocking, done-check failure and success, no secrets in artifacts, and workflow safety boundaries.

Definition of done:
A nontechnical operator can choose an example mission, run one GitHub Action, receive a complete proof-backed governed decision state, review a PR, and publish only after human approval.

GitHub Actions Operator Runbook

How to run GoalOS Mission OS autonomously until evidence is done while keeping publication human-gated.

Operator path

A nontechnical operator should be able to choose an example mission, run one workflow, review one PR, and publish only after human approval.

```
1. Copy the technical kit into the repo.
2. Commit the files on a feature branch.
3. Open GitHub Actions.
4. Run goalos-mission-os-until-done.yml with an example mission.
5. Download artifacts or review the generated PR.
6. Inspect Evidence Docket, Decision State, Claim Boundary, and QA Report.
7. Merge only after human review.
```

Until-done loop

```
MISSION_CREATED -> COMMIT_READY -> PLAN_READY -> RESEARCH_READY -> CLAIMS_READY -> VERIFICATION_READY -> DOCKET_READY ->
DECISION_STATE_READY -> ACTION_GRAPH_READY -> CHRONICLE_READY -> QA_PASSED -> HUMAN_REVIEW_READY -> DONE
```

The loop repairs missing or failed artifacts until the done checker passes. It should never auto-merge, broadcast mainnet transactions, move tokens, or promote unsupported public claims.

Required generated artifacts

Artifact	Purpose
GoalOSCommit.md	mission authority and constraints
RunCommitment.md	execution plan and work graph
ClaimsMatrix.csv	claim-to-evidence accountability
SourceProvenance.csv	source and freshness tracking
ContradictionRegister.md	uncertainty and conflict record
EvidenceDocket.md	public-safe proof room
VerifierReport.md	verification status and blocked claims
RiskLedger.csv	risks, likelihood, impact, mitigation
DecisionState.json	governed decision object

ActionGraph.md	next actions and rollback conditions
ChronicleEntry.md	reusable institutional memory

Safety boundary

- No auto-merge.
- No Mainnet broadcast.
- No token movement.
- No public claim promotion without human review.
- No production release.
- No achieved AGI/ASI, ROI, legal, financial, medical, audit, security, or mainnet claims unless evidence exists.

GOALOS MISSION OS

Launch Copy, Sales, and Pricing

Copy-paste commercial language for the first public Mission OS launch.

Homepage hero

GoalOS Mission OS — Set the objective. GoalOS runs until proof is done.

Turn high-stakes objectives into proof-backed governed decision states: Evidence Dockets, verifier reports, action graphs, risk ledgers, executive briefs, and Chronicle memory.

```
Primary CTA: Launch a Mission
Secondary CTA: View a Proof Bundle
Trust chips: Runs until done | Proof-backed outputs | Human-governed | Private by design
```

Launch post

```
Introducing GoalOS Mission OS.

The next frontier is not autonomous research.
It is autonomous proof-backed work.

A report is not enough for high-stakes decisions. Institutions need evidence, source provenance, contradiction checks,
verifier reports, risk ledgers, action graphs, rollback conditions, and memory of what proved useful.

GoalOS turns one objective into a governed decision state:
Objective -> Mission Contract -> Autonomous Work -> Verification -> Evidence Docket -> Decision State -> Action Graph ->
Chronicle.

AI creates output. GoalOS creates proof.
Set the objective. GoalOS runs until proof is done.
```

Sales email

```
Subject: From AI reports to proof-backed missions

Most AI tools produce documents.

GoalOS Mission OS produces governed decision states.

Give GoalOS one high-stakes objective. It returns an Evidence Docket, source provenance map, contradiction register,
verifier report, risk ledger, action graph, executive brief, decision deck, and Chronicle entry.

The goal is not more text.
The goal is a decision you can inspect, defend, and act on.
```

```
AI creates output. GoalOS creates proof.

We are opening the first five founding proof missions.
```

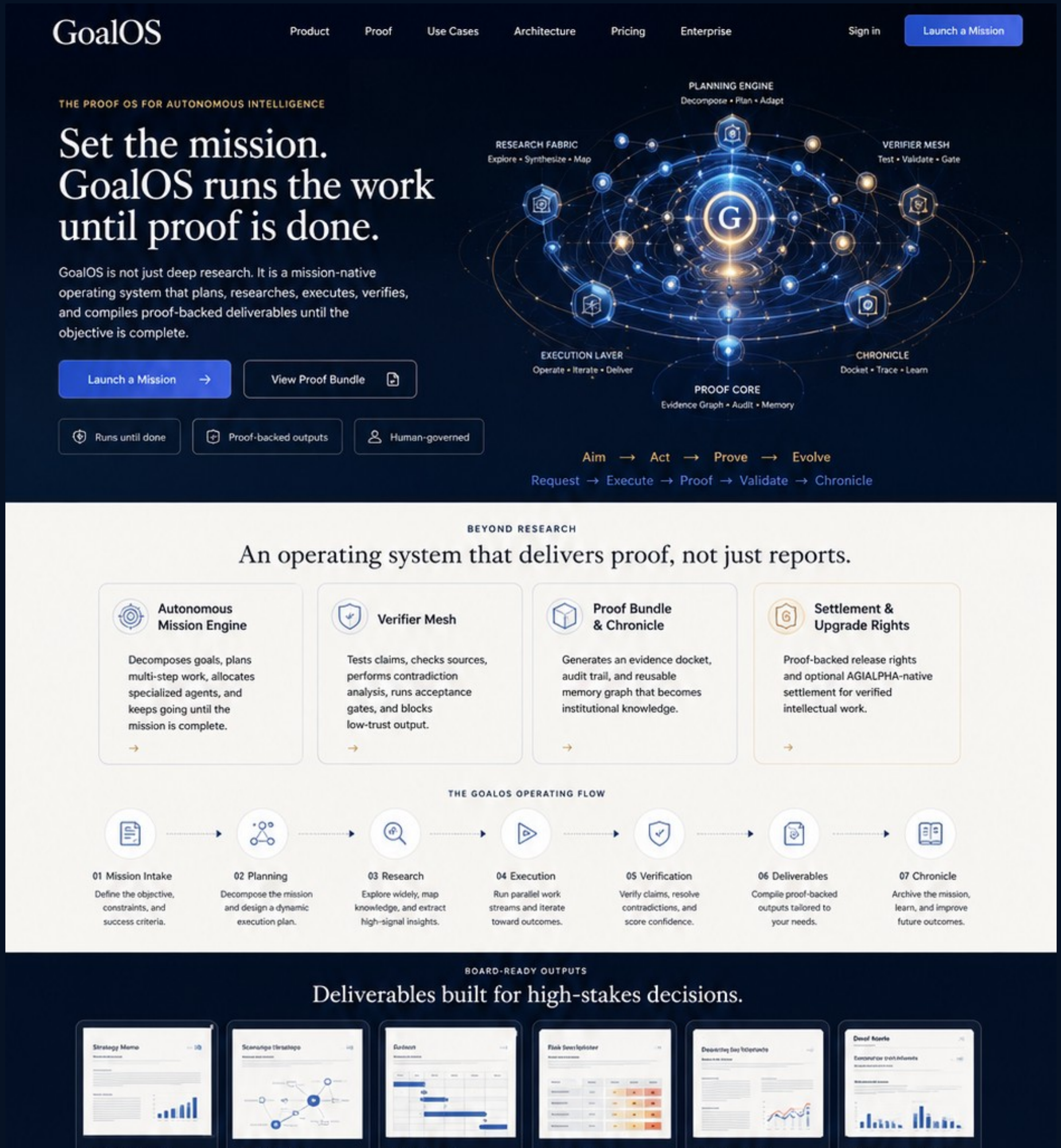
Pricing language

Package	Suggested price	Includes
Starter Mission	$2,500-$7,500	48-72 hour proof-backed decision package, public sources, Evidence Docket, Execu
Board Mission	$15,000-$50,000	Board-ready package, scenario map, risk ledger, verifier report, decision deck, priv
Enterprise Mission OS	$10,000-$100,000/month	Multiple missions, Evidence Vault, reviewer workflow, custom verifier mesh.
Sovereign / Strategic	Custom	Private deployment, institutional Evidence Dockets, governed workflows, settlemer

GoalOS Mission OS

Set the objective. GoalOS runs until proof is done.

AI creates output. GoalOS creates proof.
No proof, no settlement. No eval, no propagation. No rollback, no release.



The market has a missing layer

AI can generate output. Institutions need proof-backed decision states.

Reports inform

They summarize information but often leave claims, contradictions, risks, and action ungoverned.

Decision states govern

They define what is known, what remains uncertain, what action follows, and what proof is attached.

Evidence compounds

Chronicle memory turns validated work into reusable capability.

The new primitive

Governed Decision State = evidence + verification + risk + action + Chronicle.

Mission Contract

Designed for high-stakes autonomous AI work.

Evidence Docket

Designed for high-stakes autonomous AI work.

Verifier Mesh

Designed for high-stakes autonomous AI work.

Action Graph

Designed for high-stakes autonomous AI work.

Chronicle

Designed for high-stakes autonomous AI work.

Capability Package

Designed for high-stakes autonomous AI work.

The GoalOS flow

Objective -> Mission Contract -> Autonomous Work -> Verification -> Evidence Docket -> Decision State -> Action Graph -> Chronicle.

Objective 1	**Contract** 2	**Work** 3	**Verify** 4	**Docket** 5	**Decision** 6	**Action** 7	**Chronicle** 8

Board-ready deliverables

Evidence Docket, Verifier Report, Risk Ledger, Action Graph, Executive Brief, Decision Deck, Capability Package.

Proof-backed outputs
Designed for high-stakes autonomous AI work.

Human-governed
Designed for high-stakes autonomous AI work.

Private by design
Designed for high-stakes autonomous AI work.

Runs until done
Designed for high-stakes autonomous AI work.

Claim-bounded
Designed for high-stakes autonomous AI work.

Reusable capability
Designed for high-stakes autonomous AI work.

Why GoalOS wins

The moat is the evidence graph. Autonomy is earned. No proof, no propagation.

Proof-backed outputs
Designed for high-stakes autonomous AI work.

Human-governed
Designed for high-stakes autonomous AI work.

Private by design
Designed for high-stakes autonomous AI work.

Runs until done
Designed for high-stakes autonomous AI work.

Claim-bounded
Designed for high-stakes autonomous AI work.

Reusable capability
Designed for high-stakes autonomous AI work.

Commercial offer

Five founding proof missions: 48-hour proof-backed decision package.

Starter Mission

$2,500-$7,500
48-72 hour proof-backed package.

Board Mission

$15K-$50K
Board-ready decision state.

Enterprise Mission OS

$10K-$100K/month
Evidence Vault + workflow.

Autonomous until done

Generate -> check -> repair -> docket -> QA -> human review ready -> DONE.

Proof-backed outputs
Designed for high-stakes autonomous AI work.

Human-governed
Designed for high-stakes autonomous AI work.

Private by design
Designed for high-stakes autonomous AI work.

Runs until done
Designed for high-stakes autonomous AI work.

Claim-bounded
Designed for high-stakes autonomous AI work.

Reusable capability
Designed for high-stakes autonomous AI work.

Launch now

Publish /mission-os, run three proof missions, open paid pilots.

Launch page
/mission-os with mission intake and proof bundle sample.

Proof missions
Run three public-safe missions and open five founding pilots.

GOALOS MISSION OS

Set the objective. GoalOS runs until proof is done.

The Proof OS for autonomous AI work.

AI creates output. GoalOS creates proof.

No proof, no settlement. No rollback, no release.

Governed Decision State

Evidence Docket • Verifier Report • Risk Ledger • Action Graph • Chronicle

The moat is the evidence graph.

A living record of missions, proofs, reviews, decisions, and reusable capability.

From topic to mission.

From report to proof.

From insight to action.

Launch a Mission → View a Proof Bundle